

# Chemistry

### SECTION 1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the
- correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme:**  
*Full Marks* : +3 If **ONLY** the correct option is chosen;  
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered);  
*Negative Marks* : -1 In all other cases.

**\*Q.1** During sodium nitroprusside test of sulphide ion in an aqueous solution, one of the ligands coordinated to the metal ion is converted to

- (A)  $\text{NOS}^-$  (B)  $\text{SCN}^-$   
(C)  $\text{SNO}^-$  (D)  $\text{NCS}^-$

**Ans. A**

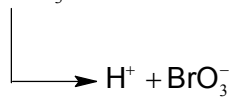
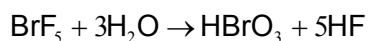
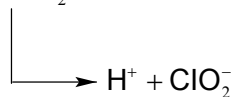
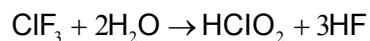
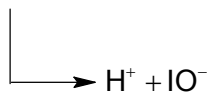
**Sol.**  $\text{S}^{2-} + [\text{Fe}(\text{CN})_5\text{NO}]^{2-} \rightarrow [\text{Fe}(\text{CN})_5(\text{NOS})]^{4-}$   
(violet)

**Q.2** The complete hydrolysis of  $\text{ICl}$ ,  $\text{ClF}_3$  and  $\text{BrF}_5$ , respectively, gives

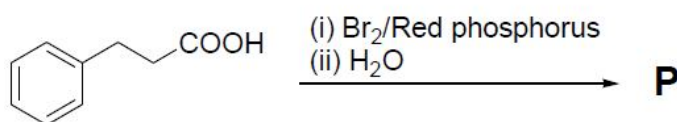
- (A)  $\text{IO}^-$ ,  $\text{ClO}_2^-$  and  $\text{BrO}_3^-$  (B)  $\text{IO}_3^-$ ,  $\text{ClO}_2^-$  and  $\text{BrO}_3^-$   
(C)  $\text{IO}^-$ ,  $\text{ClO}^-$  and  $\text{BrO}_2^-$  (D)  $\text{IO}_3^-$ ,  $\text{ClO}_4^-$  and  $\text{BrO}_2^-$

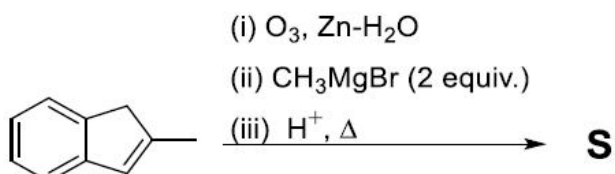
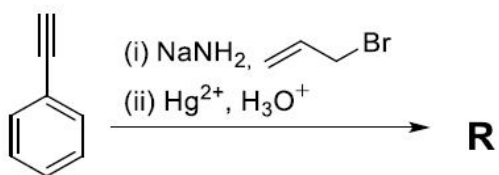
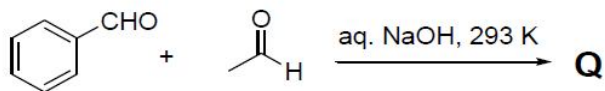
**Ans. A**

**Sol.**  $\text{ICl} + \text{H}_2\text{O} \rightarrow \text{HOI} + \text{HCl}$



**Q.3** Monocyclic compounds **P**, **Q**, **R** and **S** are the major products formed in the reaction sequences given below.

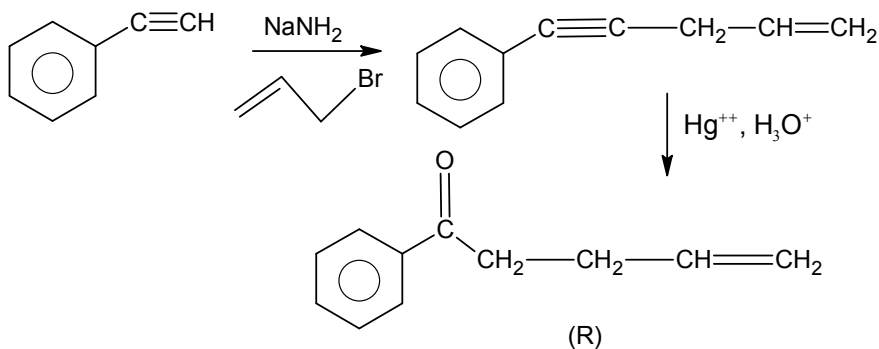
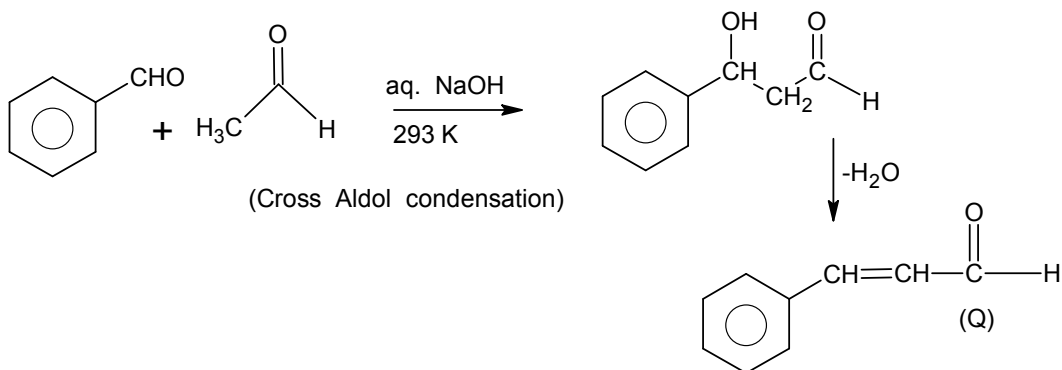
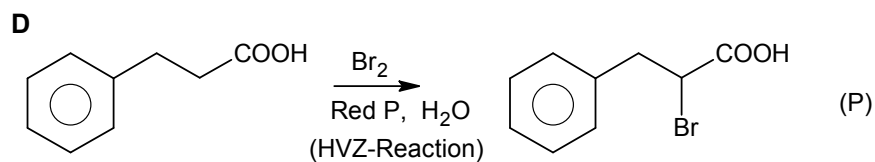


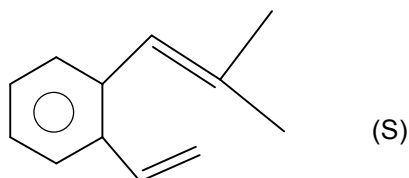
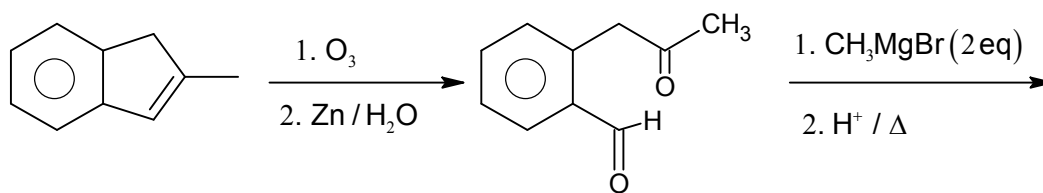


The product having the highest number of unsaturated carbon atom(s) is

- (A) **P** (B) **Q**  
(C) **R** (D) **S**

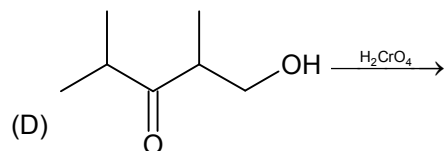
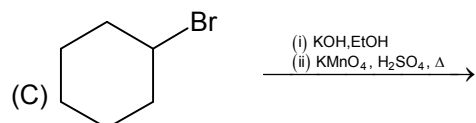
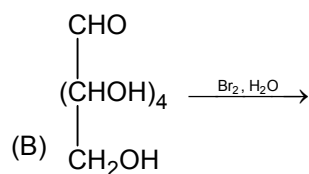
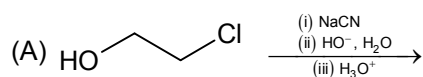
**Ans.**  
**Sol.**



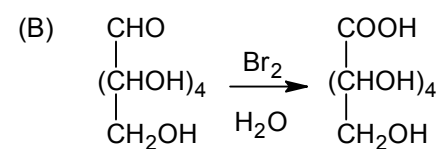
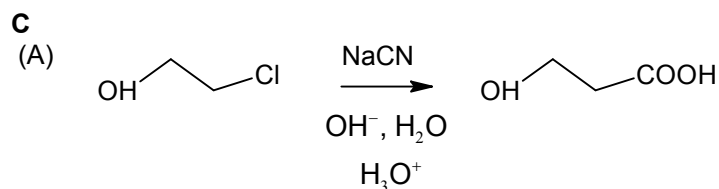


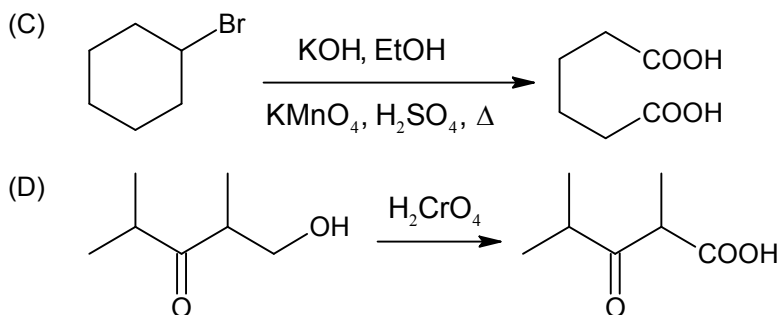
P  $\equiv$  No. of unsaturated carbon(s) = 6  
 Q  $\equiv$  No. of unsaturated carbon(s) = 8  
 R  $\equiv$  No. of unsaturated carbon(s) = 8  
 S  $\equiv$  No. of unsaturated carbon(s) = 10

**Q.4** The correct reaction/reaction sequence that would produce a dicarboxylic acid as the major product is



**Ans.  
Sol.**



**SECTION 2 (Maximum Marks: 16)**

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme**:  
*Full Marks* : +4 **ONLY** if (all) the correct option(s) is(are) chosen;  
*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen;  
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;  
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;  
*Zero Marks* : 0 If none of the option is chosen (i.e. the question is unanswered);
- Negative Marks* : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then  
 choosing **ONLY** (A), (B) and (D) will get +4 marks;  
 choosing **ONLY** (A) and (B) will get +2 marks;  
 choosing **ONLY** (A) and (D) will get +2 marks;  
 choosing **ONLY** (B) and (D) will get +2 marks;  
 choosing **ONLY** (A) will get +1 mark;  
 choosing **ONLY** (B) will get +1 mark;  
 choosing **ONLY** (D) will get +1 mark;  
 choosing no option(s) (i.e. the question is unanswered) will get 0 marks and  
 choosing any other combination of option(s) will get -2 marks.

- \*Q.5** The correct statement(s) about intermolecular forces is(are)
- (A) The potential energy between two point charges approaches zero more rapidly than the potential energy between a point dipole and a point charge as the distance between them approaches infinity.
- (B) The average potential energy of two rotating polar molecules that are separated by a distance  $r$  has  $1/r^3$  dependence.
- (C) The dipole-induced dipole average interaction energy is independent of temperature.
- (D) Nonpolar molecules attract one another even though neither has a permanent dipole moment.

**Ans. D or C, D**

**Sol.** (A) Ion – Ion  $\propto \frac{1}{r}$   
 Ion – dipole  $\propto \frac{1}{r^2}$

(B)  $F \propto \frac{1}{r^6}$  (For rotating polar molecules)

(C) As  $T \uparrow \Rightarrow \text{Energy} \downarrow$

(D) London forces or dispersion forces are the force of attraction between the non-polar molecules.

**Q.6** The compound(s) with P – H bond(s) is(are)

(A)  $\text{H}_3\text{PO}_4$

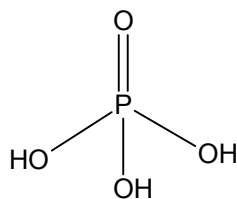
(B)  $\text{H}_3\text{PO}_3$

(C)  $\text{H}_4\text{P}_2\text{O}_7$

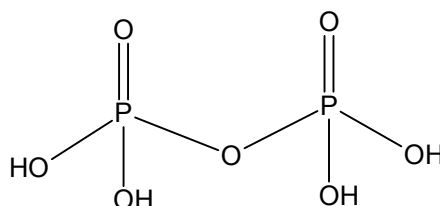
(D)  $\text{H}_3\text{PO}_2$

**Ans. B, D**

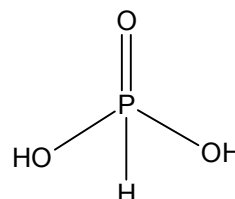
**Sol. (A)**



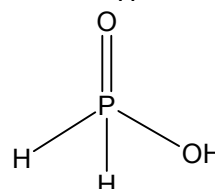
(C)



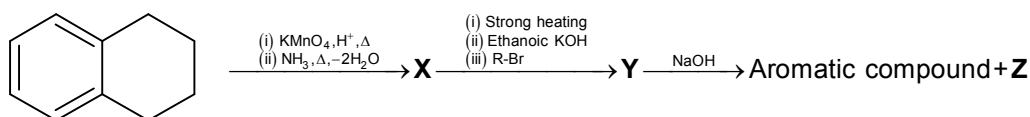
(B)



(D)



**Q.7** For the reaction sequence given below, the correct statement(s) is(are)



(A) Both **X** and **Y** are oxygen containing compounds.

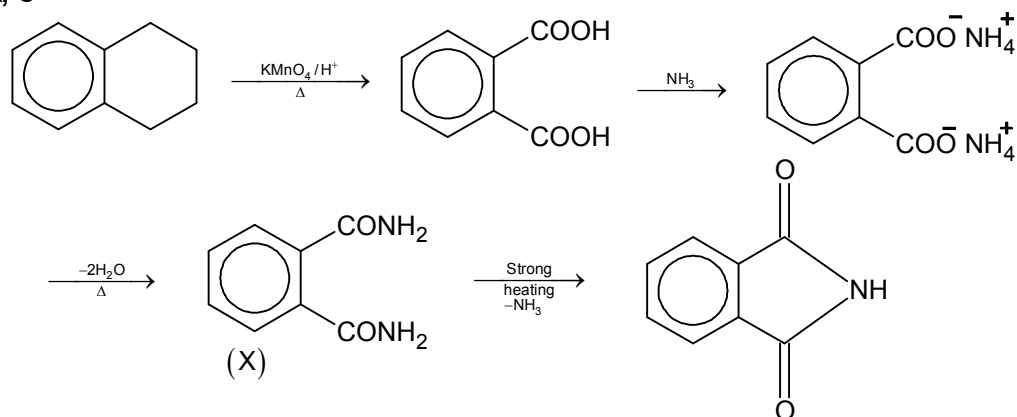
(B) **Y** on heating with  $\text{CHCl}_3/\text{KOH}$  forms isocyanide.

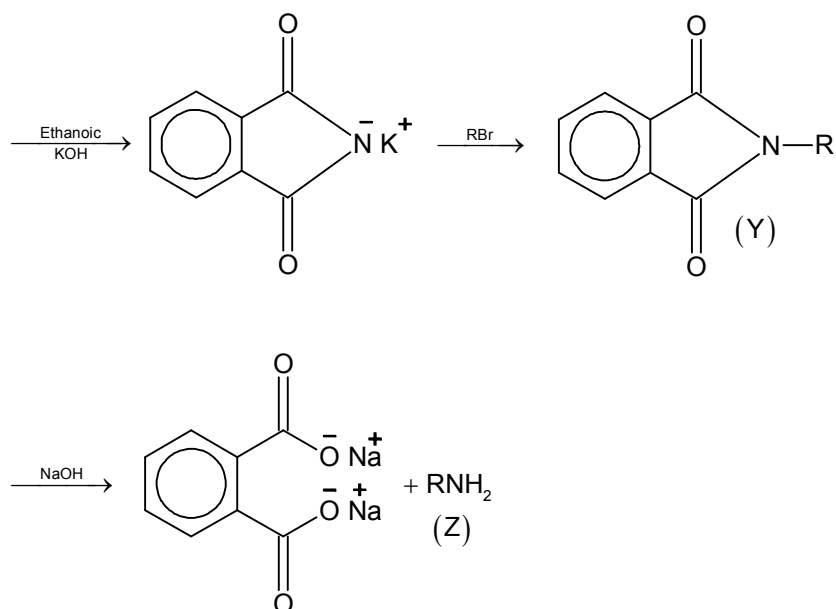
(C) **Z** reacts with Hinsberg's reagent.

(D) **Z** is an aromatic primary amine.

**Ans. A, C**

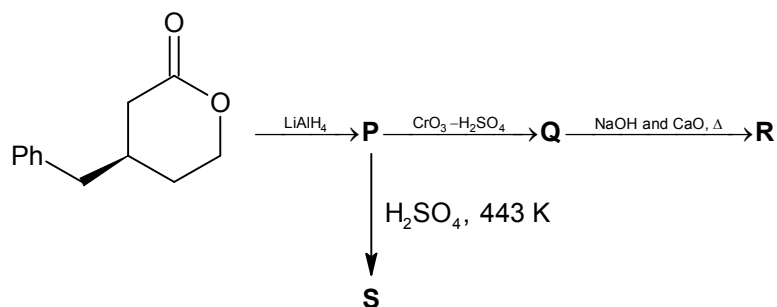
**Sol.**





(X, Y) – Oxygen containing compounds.  
 (Z) – 1° amine, so reacts with Hinsberg's reagent.

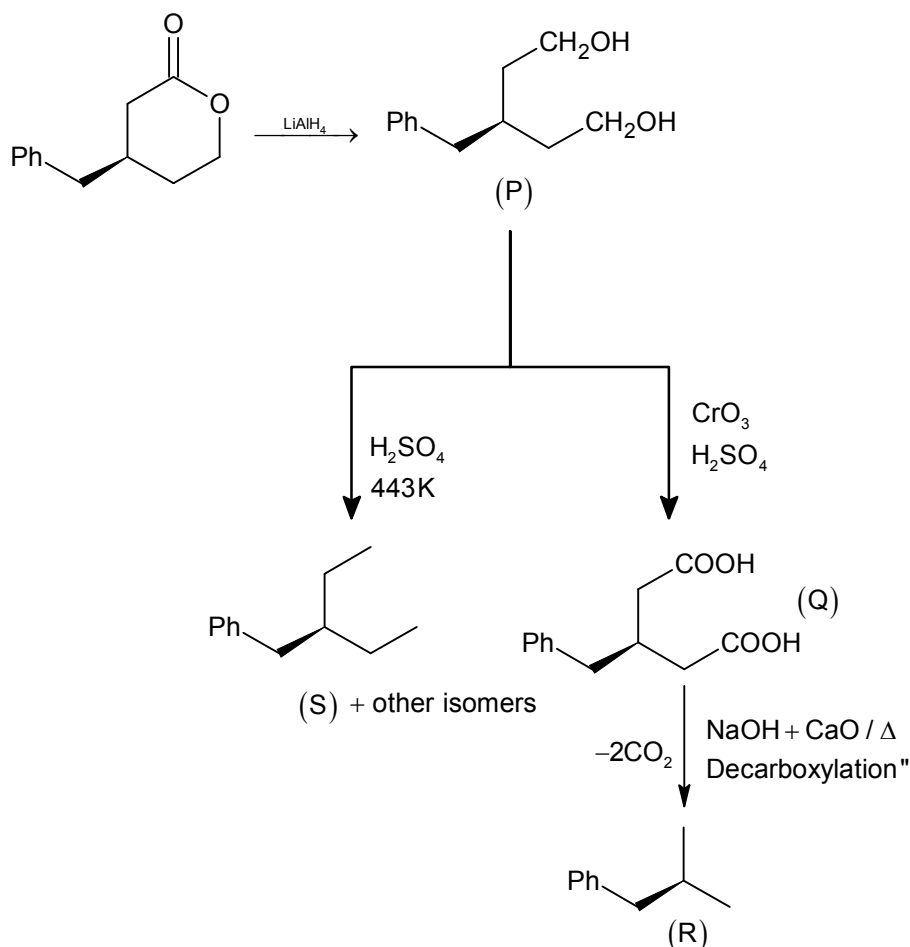
**Q.8** For the reaction sequence given below, the correct statement(s) is(are)



- (A) **P** is optically active.  
 (B) **S** gives Bayer's test.  
 (C) **Q** gives effervescence with aq. NaHCO<sub>3</sub>.  
 (D) **R** is an alkyne.

**Ans. B, C**

Sol.



P is optically inactive.

S gives Bayer's test

Q gives effervescences with  $\text{NaHCO}_3$  (aq)

R is an Alkane.

**SECTION 3 (Maximum Marks: 32)**

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme**:  
*Full Marks* : +4 If **ONLY** the correct numerical value is entered in the designated place;  
*Zero Marks* : 0 In all other cases.

- Q. 9** The density (in  $\text{g cm}^{-3}$ ) of the metal which forms a cubic close packed (ccp) lattice with an axial distance (edge length) equal to  $400\text{ pm}$  is \_\_\_\_\_.  
**Use:** Atomic mass of metal =  $105.6\text{ amu}$  and Avogadro's constant =  $6 \times 10^{23}\text{ mol}^{-1}$

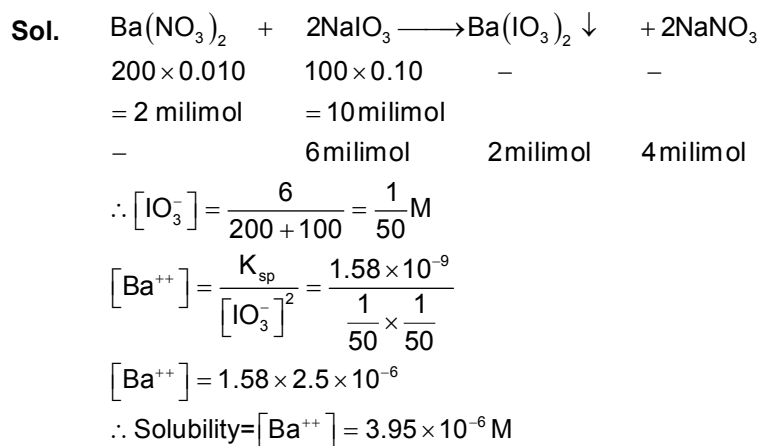
Ans. 11.00

**Sol.** For ccp,  $Z = 4$ ,  $M = 105.6 \text{ g mol}^{-1}$   
 $N_A = 6 \times 10^{23} \text{ mol}^{-1}$   
 $a = 400 \text{ pm} = 4 \times 10^{-8} \text{ cm}$   
 $\therefore d = \frac{ZM}{a^3 N_A} = \frac{4 \times 105.6}{(4 \times 10^{-8})^3 \times 6 \times 10^{23}}$   
 $d = \frac{4 \times 105.6}{64 \times 6 \times 10^{-24} \times 10^{23}}$   
 $d = \frac{1056}{96} = 11 \text{ cm}^{-3}$

**\*Q. 10** The solubility of barium iodate in an aqueous solution prepared by mixing 200 mL of 0.010 M barium nitrate with 100 mL of 0.10 M sodium iodate is  $X \times 10^{-6} \text{ mol dm}^{-3}$ . The value of  $X$  is \_\_\_\_\_.

**Use:** Solubility product constant ( $K_{sp}$ ) of barium iodate =  $1.58 \times 10^{-9}$ .

**Ans. 3.95**



**Q. 11** Adsorption of phenol from its aqueous solution on to fly ash obeys Freundlich isotherm. At a given temperature, from  $10 \text{ mg g}^{-1}$  and  $16 \text{ mg g}^{-1}$  aqueous phenol solutions, the concentrations of adsorbed phenol are measured to be  $4 \text{ mg g}^{-1}$  and  $10 \text{ mg g}^{-1}$ , respectively. At this temperature, the concentration (in  $\text{mg g}^{-1}$ ) of adsorbed phenol from  $20 \text{ mg g}^{-1}$  aqueous solution of phenol will be \_\_\_\_\_.

**Use:**  $\log_{10} 2 = 0.3$

**Ans. 16.00**

**Sol.**  $\frac{x}{m} = KC^{1/n}$

$C = 10 \text{ mg g}^{-1} \Rightarrow \frac{x}{m} = 4 \text{ mg g}^{-1}$

$C = 16 \text{ mg g}^{-1} \Rightarrow \frac{x}{m} = 10 \text{ mg g}^{-1}$

$\Rightarrow 4 = K(10)^{1/n} \quad (1)$

$10 = K(16)^{1/n} \quad (2)$

$\Rightarrow \frac{4}{10} = \left(\frac{10}{16}\right)^{1/n}$

$$\Rightarrow \log 4 - \log 10 = \frac{1}{n} [\log 10 - \log 16]$$

$$\Rightarrow n = \frac{1 - 4 \times .30}{2 \times .30 - 1} = \frac{0.2}{0.4} \Rightarrow n = \frac{1}{2}$$

$$\text{Now } \frac{x}{m} = K (20)^{\frac{1}{x}} \quad (3)$$

Dividing (3) by (1)

$$\frac{x/m}{4} = \left( \frac{20}{10} \right)^{\frac{1}{x}} = 4$$

$$\frac{x}{m} = 16 \text{ mg g}^{-1}$$

- Q. 12** Consider a reaction  $A + R \longrightarrow \text{Product}$ . The rate of this reaction is measured to be  $k[A][R]$ . At the start of the reaction, the concentration of  $R$ ,  $[R]_0$ , is 10-times the concentration of  $A$ ,  $[A]_0$ . The reaction can be considered to be a pseudo first order reaction with assumption that  $k[R] = k'$  is constant. Due to this assumption, the relative error (in %) in the rate when this reaction is 40% complete, is \_\_\_\_\_.  
[ $k$  and  $k'$  represent corresponding rate constants]

**Ans.** 4.17

**Sol.** Let  $[A]_0 = 1 \text{ M}$ ,  $[R]_0 = 10 \text{ M}$   
Considering pseudo first order reaction  
 $r = k'[A]$ , on 40% completion  
or  $r_1 = k[10][0.6] \Rightarrow r_1 = 6K \quad \dots (1)$   
Considering second order reaction  
 $r = k[R][A]$ , on 40% completion  
 $r_2 = k[9.6][0.6] \Rightarrow r_2 = 5.76 K \quad \dots (2)$   
 $\therefore$  From (1) and (2)  
Relative error =  $\frac{6K - 5.76K}{5.76K} \times 100$   
 $= \frac{.24}{5.76} \times 100 = \frac{100}{24} = 4.166$   
 $\cong 4.17\%$

- Q. 13** At 300 K, an ideal dilute solution of a macromolecule exerts osmotic pressure that is expressed in terms of the height ( $h$ ) of the solution (density =  $1.00 \text{ g cm}^{-3}$ ) where  $h$  is equal to 2.00 cm. If the concentration of the dilute solution of the macromolecule is  $2.00 \text{ g dm}^{-3}$ , the molar mass of the macromolecule is calculated to be  $X \times 10^4 \text{ g mol}^{-1}$ . The value of  $X$  is \_\_\_\_\_.  
**Use:** Universal gas constant ( $R$ ) =  $8.3 \text{ J K}^{-1} \text{ mol}^{-1}$  and acceleration due to gravity ( $g$ ) =  $10 \text{ m s}^{-2}$ .

**Ans.** 2.49

**Sol.** Osmotic pressure  $\Pi = h \rho g$   
 $= 2 \times 10^{-2} \text{ m} \times \frac{1 \text{ kg}}{\text{m}^3} \times \frac{10 \text{ m}}{\text{s}^2}$   
 $= 0.2 \text{ Pa}$

Now  $0.2 = CRT$

$$\Rightarrow 0.2 = 8.3 \times 300 \times \frac{2}{M}$$

$$\Rightarrow M = 8.3 \times 3000$$

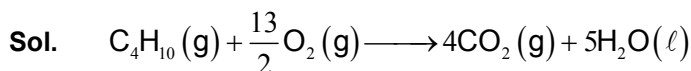
$$\Rightarrow M = 24900.0 = 2.49 \times 10^4 \text{ g mol}^{-1}$$

- Q. 14** An electrochemical cell is fueled by the combustion of butane at 1 bar and 298 K. Its cell potential is  $\frac{X}{F} \times 10^3$  volts, where  $F$  is the Faraday constant. The value of  $X$  is \_\_\_\_\_.

**Use:** Standard Gibbs energies of formation at 298 K are :

$$\Delta_f G^\circ_{\text{CO}_2} = -394 \text{ kJ mol}^{-1}; \Delta_f G^\circ_{\text{water}} = -237 \text{ kJ mol}^{-1}; \Delta_f G^\circ_{\text{butane}} = -18 \text{ kJ mol}^{-1}$$

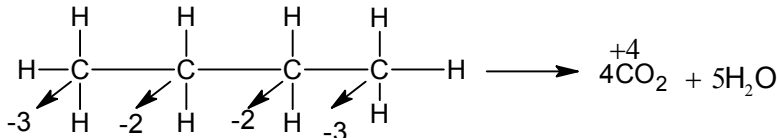
**Ans. 105.50**



$$\Delta_r G^\circ = 4 \times (-394) + 5 \times (-237) - (1 \times -18)$$

$$\Delta_r G^\circ = -1576 - 1185 + 18$$

$$\Delta_r G^\circ = -1558 - 1185 = -2743 \text{ KJ mol}^{-1}$$



No. of  $e^-$ (s) exchanged (lost)

$$= 2 \times [4 - (-3)] + 2 \times [4 - (-2)]$$

$$= 14 + 12 = 26$$

$$\Delta G = -nF E_{\text{cell}}$$

$$-2743 \times 10^3 = -26 \times F \times E_{\text{cell}}$$

$$E_{\text{cell}} = \frac{2743}{26F} \times 10^3 = \frac{105.5}{F} \times 10^3$$

$$X = 105.50$$

- Q. 15** The sum of the spin only magnetic moment values (in B.M.) of  $[\text{Mn}(\text{Br})_6]^{3-}$  and  $[\text{Mn}(\text{CN})_6]^{3-}$  is \_\_\_\_\_.

**Ans. 7.72**

**Sol.**  $[\text{MnBr}_6]^{3-} \Rightarrow \text{Mn}^{3+} \Rightarrow 3d^4 \Rightarrow n = 4$ , since  $\text{Br}^-$  is a weak field ligand.

$[\text{Mn}(\text{CN})_6]^{3-} \Rightarrow \text{Mn}^{3+} \Rightarrow 3d^4 \Rightarrow n = 2$ , since  $\text{CN}^-$  is a strong field ligand.

$$\therefore \mu_s(\text{total}) = \sqrt{4(4+2)} + \sqrt{2(2+2)}$$

$$= \sqrt{24} + \sqrt{8}$$

$$= 4.90 + 2.82$$

$$\therefore \mu_s(\text{total}) = 7.72 \text{ BM}$$

- Q. 16** A linear octasaccharide (molar mass =  $1024 \text{ g mol}^{-1}$ ) on complete hydrolysis produces three monosaccharides: ribose, 2-deoxyribose and glucose. The amount of 2-deoxyribose formed is 58.26% (w/w) of the total amount of the monosaccharides produced in the hydrolyzed products. The number of ribose unit(s) present in one molecule of octasaccharide is \_\_\_\_\_.  
**Use:** Molar mass (in  $\text{g mol}^{-1}$ ): ribose = 150, 2-deoxyribose = 134, glucose = 180; Atomic mass (in amu) : H = 1, O = 16

**Ans. 2.00**

**Sol.**

